

Components of Developing a Point of Sale (POS) System for Student Projects

1. User Interface (Front-End)

This interface allows cashiers or administrators to interact with the POS system.

Key features: product search, barcode input, shopping cart display, quantity adjustment, discount application, checkout screen, and receipt preview.

Possible technologies: Python (Tkinter/PyQt), Java (JavaFX), C# (.NET), or Web technologies (HTML, CSS, JavaScript).

2. Product Management Module

Allows storage and management of products sold in the system.

Functions include adding, editing, deleting products, categorization, pricing, and barcode management.

Typical fields: Product ID, Product Name, Category, Price, Stock Quantity, Supplier.

3. Inventory Management System

Tracks stock levels in real time.

Functions include automatic stock deduction after sale, low stock alerts, stock replenishment, stock reports, and inventory adjustments.

4. Sales Processing Module

Handles sales transactions in the POS.

Processes include creating a sale, adding items to cart, calculating totals, applying taxes or discounts, and generating receipts.

5. Payment Processing Module

Supports different payment methods such as cash, mobile money, card payments, or split payments.

Functions include payment confirmation, change calculation, and payment record storage.

6. Customer Management Module

Stores customer information and purchase history.

Typical fields: Customer ID, Name, Phone Number, Email, Address, and Loyalty Points.

7. Database System

The database stores all POS data.

Typical tables: Users, Products, Sales, Sales Items, Customers, Inventory, and Payments.

Common databases: MySQL, PostgreSQL, SQLite, SQL Server.

8. Reporting and Analytics

Provides reports to monitor business performance.

Examples: Daily sales reports, product sales reports, profit reports, inventory reports, and cashier performance reports.

9. User Authentication and Role Management

Ensures system security.

Roles include Admin, Manager, and Cashier.

Features include login authentication, password encryption, and access control.

10. Receipt Generation

The system automatically generates receipts.

Receipt details include store name, date/time, purchased items, taxes, total amount, and payment method.

11. Hardware Integration

Optional advanced component integrating devices such as barcode scanners, receipt printers, cash drawers, and card readers.

12. Backup and Data Recovery

Ensures system reliability through database backup, restore functionality, and data export.

Architecture

- **System Architecture**
- Students should design the POS system using a three-tier architecture.
- Architecture Layers:
- Presentation Layer (User Interface)
 - Cashier screen
 - Admin dashboard
 - Product search interface
 - Checkout page
- Application Layer (Business Logic)
 - Sales processing
 - Inventory control
 - Payment handling
 - Report generation
- Data Layer (Database)
 - Storage of products
 - Sales transactions
 - Customer records
 - Inventory data

- **Main POS Modules**
- Module 1: Authentication System
- Features:
 - Login
 - Logout
 - Password encryption
 - Role-based access
- Roles:
 - Administrator
 - Manager
 - Cashier
- Module 2: Product Management System
- Functions:
 - Add product
 - Update product
 - Delete product
 - Search product
 - View product list
- Typical Product Fields:
 - Product ID
 - Product name
 - Category
 - Price
 - Quantity
 - Barcode
- Module 3: Inventory Management
- Functions:
 - Update stock after sale
 - Low stock alert
 - Stock adjustment
 - Supplier restocking
- Module 4: Sales Processing System
- Processes:
 - Scan product barcode
 - Retrieve product information
 - Add product to cart
 - Calculate total cost

- Apply discounts
- Confirm payment
- Save transaction

- Module 5: Payment Processing
- Supported payments:
 - Cash
 - Mobile money
 - Credit/debit card

- Module 6: Customer Management
- Functions:
 - Register customers
 - Track purchase history
 - Loyalty points system

- Module 7: Receipt Generation
- Receipt includes:
 - Store name
 - Transaction ID
 - Date and time
 - Purchased items
 - Prices
 - Total amount
 - Payment method

- Module 8: Reporting and Analytics
- Reports:
 - Daily sales report
 - Weekly sales report
 - Product performance report
 - Inventory report
 - Cashier sales report

- **POS Database Design**
- Students should create a relational database schema.

- Main Tables:
 - Users
 - Products
 - Customers
 - Sales

- Sales_Items
 - Inventory
 - Payments
- Example Tables:
- Products
 - product_id
 - product_name
 - category
 - price
 - quantity
 - barcode
- Sales
 - sale_id
 - date
 - user_id
 - customer_id
 - total_amount
 - payment_method
- Sales_Items
 - sale_item_id
 - sale_id
 - product_id
 - quantity
 - price
- **Hardware Components (Optional)**
- Possible POS hardware integration:
 - Barcode scanner
 - Receipt printer
 - Cash drawer
 - Card reader
 - Touchscreen display
- **Security Features**
 - Password hashing
 - Role-based access control
 - Transaction logs

- Database backup

- **Technologies Students Can Use**

- Option 1: Desktop POS
 - Python
 - Tkinter
 - SQLite
- Option 2: Web POS
 - HTML / CSS / JavaScript
 - PHP / Django / Node.js
 - MySQL / PostgreSQL
- Option 3: Enterprise POS
 - Java + Spring Boot
 - React frontend
 - PostgreSQL